

When My Car Whispers Secrets to My Home: Envisioning the Opportunities and Challenges of Human-Vehicle-Home Integrated Ecosystems through Probes

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Abstract

This paper explores the concept of human-vehicle-home (HVH) integrated intelligent experiences and its potential to transform everyday life through seamless connectivity between smart vehicles and smart homes. While existing research has largely focused on technical advancements, this study adopts a human-centered approach to uncover the design opportunities and socio-technical challenges inherent in HVH systems. Using a probe-based method, we engaged eight participants from diverse living conditions across China to envision future scenarios of HVH integration, capturing their expectations, concerns, and values. Our findings highlight the potential of HVH systems in enhancing cross-space well-being, fostering continuums in activity and memory, and strengthening interpersonal support, particularly within families. However, participants also identified critical challenges, including privacy risks, safety concerns, mismatches in technological literacy, and usability disparities across regions. This work provides the first exploration of HVH systems and design implications that move beyond technical advancement to prioritize human needs and values, as well as embrace the diversity of communities and lifestyles.

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CCS Concepts

• **Human-centered computing** → **Participatory design; User studies.**

Keywords

Smart Environment, Human-Vehicle-Home Integration, Human-centered Design, Smart Vehicle, Smart Home Technology, Privacy and Security, Participatory Design

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1 Introduction

In recent years, the convergence of the Internet of Things (IoT), Artificial Intelligence (AI), and Smart Home Technologies (SHT), along with the growing adoption of Electric Vehicles (EVs), has propelled people toward a more interconnected and integrated intelligent lifestyle. The connectivity between intelligent vehicles and homes equipped with smart devices and sensors is illuminating new visions for future smart living. For instance, people can already adjust their home's lighting and check in-home cameras while parked at a grocery store¹, or have their car automatically

¹<https://www.telematicswire.net/home-automation-with-in-vehicle-voice-assistant>

notify them of a potential security threat through Sentry Mode² while relaxing at home. We identify this trend as a narrative of “human-vehicle-home (HVH) integrated intelligent experiences,” where people virtually link with vehicles and homes—viewed as mobile and stationary nodes—within a unified ambient intelligence ecosystem [1, 8]. These advancements lay the foundation for HCI communities to envision how seamlessly connected vehicles and homes can enhance daily convenience, security, and harmony, yet this full potential remains underexplored, especially regarding the multifaceted realities of people’s everyday lives.

Existing HCI research has made strides in extending the benefits of connected in-car intelligence, addressing areas like entertainment, health interventions, and personalized in-vehicle interactions [4, 13, 18, 21]. However, the socio-technical perspectives seen in SHT research—often addressing the dynamic interactions between complex family structures and technology—remain largely absent in considering cars as shared functional spaces. This gap is particularly critical when it comes to addressing ethical risks [24, 31], avoiding the pitfalls of techno-solutionism [2, 33], and preventing marginalization [10] associated with such innovations. Furthermore, current work on vehicle-home connectivity often focuses narrowly on technical topics like power storage and conversion [3, 32] or conceptual attempts [17], perpetuating the “technology as enchantment” narrative [20] while limiting alternative imaginaries of its evolution. However, critical questions remain unanswered: Why do people need vehicle-home connectivity? What are their expectations, and what are their concerns? To address these, we advocate for actively engaging potential end-users who are the ultimate beneficiaries of vehicle-home connectivity to envision their future, in order to ensure that technological innovation responsibly integrates into diverse, dynamic, context-rich, and socio-culturally grounded lifestyles while prioritizing human value to mitigate concerns around heightened surveillance and invasiveness.

To envision the potential contexts of people’s everyday lives that uncover design opportunities and potential risks of HVH experience, we drew inspiration from previous probe-based methods [11, 14, 28] to develop a “Human-Vehicle-Home Probe package” which included tools like miniature models, a framing device, and cards for eliciting imaginatively scenarios on intelligent vehicle-home connectivity in relation to their own lives. These materials inspired participants to reflect on how interconnected intelligent vehicles and homes might benefit or disrupt their daily routines, connecting these reflections with sociocultural factors such as living arrangements, family composition, and lifestyle habits. We recruited eight participants from various regions of China through interest forms and snowballing, aiming to capture diverse insights shaped by their distinct living conditions, experiences with smart products, and attitudes toward integrated intelligence. To facilitate this, we sent them probe packs via courier service to document relevant reflections from their daily routines.

In this work, we provide pioneering preliminary insights into participants’ envisioned scenarios of smart vehicle-home integration, capturing their notions, expectations, and concerns regarding

this emerging concept. Our contributions are twofold: First, we illustrate how reflexive and situated perspectives empower participants to critically connect their often-unseen everyday lives and values with the opportunities and risks of smart vehicle-home integration. By uncovering these nuanced perspectives, our approach challenges technology-centric HVH design paradigms and promotes deeper reflection on the limitations of techno-solutionism. Second, our findings highlight the potential of HVH systems to enhance **cross-space well-being, foster continuity in activity and memory, and strengthen interpersonal support, particularly within families**. Meanwhile, participants identified critical challenges, including **privacy risks, safety concerns, mismatches in technological literacy, and usability disparities across regions**. These insights provide actionable guidance for designing HVH systems that move beyond technical advancement to prioritize human needs and values, as well as embrace the diversity of communities and lifestyles.

2 Methodology

While engaging potential end-users in future envisioning is widely regarded as essential for advancing human-centered value in intelligent and integrated technologies, however, this process faces significant challenges with people who are not technical experts. Although we believe perspectives beyond the engineering domain are anticipated to contribute meaningfully to the agenda to advance HVH integration, previous qualitative studies [25] have identified the paradox that users’ unfamiliarity with technological frameworks undermines their confidence and critical reflection on the problematic aspects of smart narratives. Compounding this challenge is the tendency for people’s future-oriented imagination to become overly fanciful, disconnected from their everyday contexts [12, 30]. This disconnect often sidelines discussions grounded in the lived environments, routines, and values that shape people’s daily experiences. To address these challenges, we developed the Human-Vehicle-Home Probe Pack to investigate participants’ expectations and concerns about future vehicle-home connectivity. Participants from different regions of China were recruited and received the probe packs via courier services, allowing them to reflect on their daily routines and leisure activities.

2.1 Designing Human-Vehicle-Home Probe

Previous SHT research emphasizes empowering participants to connect their imagination with daily routines [14, 28]. For this purpose, probe-based design methods have been widely adopted in future-oriented HCI research, particularly in domestic contexts. Haines et al. [14] used probes like mission envelopes, cameras, and photo record books to help participants document daily values and reflect on human-centered smart home design. Similarly, Schulte et al. [28] introduced design fiction probes, encouraging participants to create speculative scenarios about future smart homes, and fostering discourse on uncertain developments. Building on these probe-based methodologies, we developed the “Human-Vehicle-Home Probe Package” to elicit narrative scenarios about idealized or problematic future vehicle-home connectivity, connecting them to participants’ real-life residential settings, daily travel routines, and interactions with other stakeholders.

²https://www.tesla.com/ownersmanual/model3/en_us/GUID-56703182-8191-4DAE-AF07-2FDC0EB64663.html

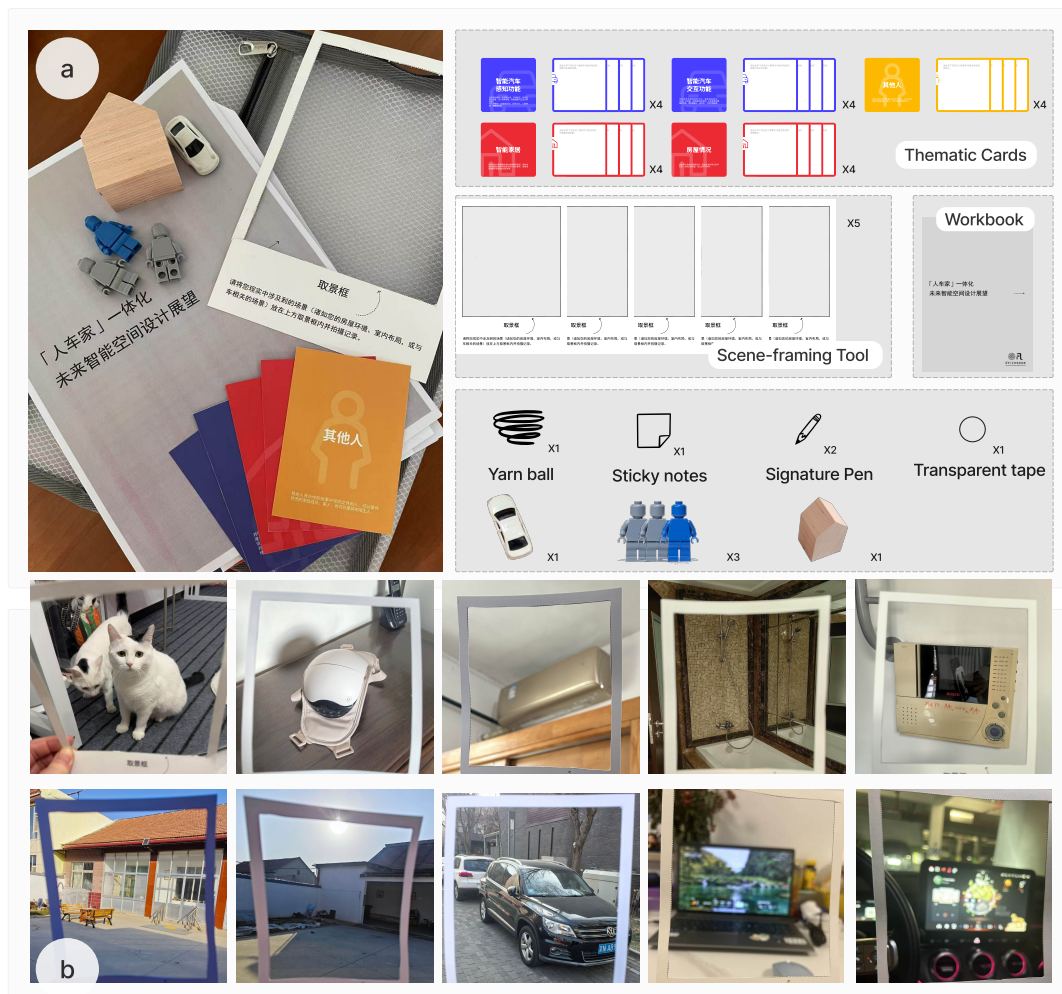


Figure 1: Design probe and its application: (A) The design probe package and tools in detail (B) Photos illustrating the usage context and environment.

2.1.1 The HVH probe package. The probe package, inspired by the cultural probe method by Gaver et al. [11], includes tools like models, cards, yarn balls and scene-framing devices, with a workbook providing guidance for tasks (See Figure 1A & Appendix). The workbook was developed as a guide to introduce the background, tasks, and usage of the provided materials to inform design envisioning. The models, cards, framing tool, and yarn were included to facilitate the generation and visualization of design fiction fragments on a canvas. Specifically, the generic models of people, cars, and living spaces—lacking distinctive features—were used to arrange spatial relationships within a scenario. The portable scene-framing tool, inspired by [35], captured rich visuals of participants’ environments while maintaining privacy with adjustable focus. The fillable thematic cards were organized into five categories, prompting participants to consider critical aspects of their imagined scenarios:

the stakeholders involved, the car's perceptual capabilities, participants' modes of interaction, environmental features (indoor and outdoor), and the integration of smart home devices. Additionally, yarn was provided to encourage participants to visualize data transmission paths within their stories, fostering reflections based on their technical understanding.

As such, the probe pack aimed to provoke participants to reflect on the design opportunities and challenges related to HVH integration in their daily lives. It also served as a "ticket to talk," facilitating conversations between participants and researchers about the narrative scenarios recorded in the workbook, the configured materials on the canvas, and the accompanying photographs.

2.2 Conducting Situated Scenario Envisioning on Human-Vehicle-Home Integration

2.2.1 Participant Recruitment. Eight participants from diverse regions and backgrounds were recruited via social media and snow-ball sampling to explore HVH connectivity. (see Table 1). This diversity provided us with a wealth of experiential knowledge and unveiled different unseen aspects of everyday life that could benefit from intelligent home-vehicle connectivity. It also offered reflective and reflexive perspectives, delivering valuable insights into design opportunities and challenges.

2.2.2 Conducting Scenario Envisioning. The study was divided into three stages: 1) **Pilot Study:** A pilot study was conducted with other researchers and co-authors from the university as participants to test the predefined procedure for scenario exploration (detailed in Stage 2). During this phase, we validated the functionality of the probe and refined the printed instructions on the thematic cards and the design workbook. 2) **Scenario Envisioning:** The probe packs were distributed to participants via courier services, with the primary task being to reflect on their daily lives and imagine how future smart vehicle-home connectivity might benefit or harm them. Participants documented their imaginative scenarios in the design workbook and used white paper as a canvas to depict key information, such as spatial relationships, potential smart services, and stakeholders. This process involved arranging annotated models, sticky notes, and sketches. Contextual connections in their daily life were captured using the scene-framing tool and phone cameras (see Figure 1B). Each participant was allocated up to four days to finish the entire process. They were finally required to create four future life scenarios involving vehicle-home connectivity: two positive examples and two negative ones (deemed concerning or risky). The textual descriptions of these scenarios in workbooks, the constructed canvases, and framing device photos, were submitted to researchers via WeChat. 3) **Follow-up Meetings:** Follow-up online meetings were conducted with all participants after an initial analysis and discussion of the submitted content, during which researchers identified themes of interest from recurring narratives, visual patterns, and participant reflections. We also examined connections between participants' identities, living habits, housing situations, and their expectations or concerns, as well as the deeper needs and considerations underlying their envisioned stories. In these meetings, participants were interviewed about their motivations behind the creation process, the stories and contexts behind specific scenarios and photos, and how these connected to their real lives. The interviews were audio-recorded for subsequent analysis.

2.2.3 Data Analysis. We collected eight submissions, including 32 imaginative scenarios, photos of the constructed canvases and 38 framed photos (examples shown in Figure 2), as well as interview transcripts with every participant. This study was approved by the University's Institutional Review Board (IRB). All participants signed informed consent forms before Stage 2. We used inductive thematic analysis [5] with four coders, including three organizers and one external researcher. In the first cycle of coding, we conducted open coding on both visual (e.g., photos, canvases) and textual data (e.g., scenarios, interview transcripts) to identify initial codes. To ensure a holistic analysis, we integrated codes from the

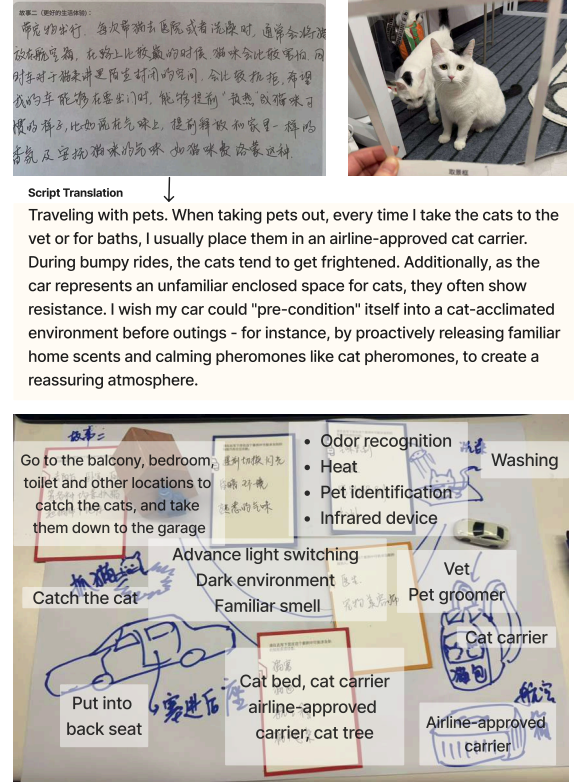


Figure 2: Examples of participants' scenario depictions with probes and visual documentation.

visual and textual data. For example, we cross-referenced participants' scenarios with their interview explanations and the visual depictions on the canvases to validate and enrich our interpretations. In the second cycle, we grouped related codes into potential themes and sub-themes, focusing on how participants used the probes to connect their imaginative scenarios with real-life details, their expectations of how HVH systems could benefit them, and their concerns about potential risks. Through two cycles of open coding on visual and textual data, we identified three key themes: scenario envisioning, expectations of HVH integration, and associated challenges.

3 Findings and Discussion

3.1 Participating Scenario Envisioning: Connecting Everyday Life, Unseen Details, and People's Values

The imaginative scenarios created by participants demonstrated strong ties between intelligent home-vehicle connectivity and various aspects of real-life experiences, such as living conditions, family structures, SHT usage, commuting routines, and broader life needs. However, it can be argued that participants were merely crafting

Table 1: Details of Participants.

ID	Gender	Housing Type	Cohabitants	Freq. of SHT Usage	Freq. of Vehicle Usage	Technical Proficiency
P1	Male	Urban Residential	Spouse and Child	Everyday	Several Times a Week	Fundamental
P2	Male	Social Housing	Spouse	Tried Only a Few Times	Everyday	Intermediate
P3	Female	Self-built House	Parents	Several Times a Week	Everyday	Advanced
P4	Female	Shared Housing	Co-housing Community	Several Times a Week	Everyday	Advanced
P5	Male	Urban Residential	Living Alone	Occasionally Each Month	Everyday	Intermediate
P6	Female	Urban Residential	Spouse and Child	Everyday	Occasionally Each Month	Fundamental
P7	Female	Rural Dwelling	Parents	Occasionally Each Month	Rarely or Almost Never	Novice
P8	Female	Urban Residential	Multi-generation	Everyday	Several Times a Week	Advanced

futuristic narratives with practical relevance. How do people subjectively construct meaningful associations between envisioned futures and their everyday surroundings? P3 illustrated a concrete example of how ideas and themes of interest are discovered in situ within his environment: “I’d keep it in mind, and when I see something—like hot water or my car—I think, ‘this could be an example.’” When participants reflected on the potential of an integrated smart ecosystem in personal terms, complex themes interweaving technology, people, environments, and cultures emerged. For instance, participants from rural areas, like P4 and P8, envisioned systems addressing regional challenges, such as automatically preheating homes as their car approaches or using cars as backup batteries and Wi-Fi transmitters during power outages. Similarly, P1, from a three-generation household in China, drew inspiration from routine family tasks, illustrating how reflective perspectives reveal integrated intelligence as both an enabler and a potential barrier in their lives. This supports “out-of-the-box thinking” [16] to HVH experience, moving beyond technology-dominated framings toward solutions that integrate contextual, situated, and socio-cultural factors. In addition, probes uncovered unseen scenarios and things related to different people’s lives, which facilitate dialogues between participants and researchers about their value, the potential opportunities and challenges in design. For example, P1 shared a scenario where the car releases a home-like scent to ease her cats’ discomfort in enclosed spaces. She highlighted that welcoming lights in her cars always terrify pets, revealing challenges in designing pet-friendly environments. Other subtle scenarios, involving objects like car interiors, yards, gates, home massagers, and smart monitoring systems, highlighted connections to participants’ lives and their deeper values around safety, privacy, and trustworthiness. Despite the dominating narrative that often portrays technology as enchantment, which can hinder people from linking what they truly value with technological solutions [7, 20], we highlight the unveiling of these unseen life fragments became the foundation for designing vehicle-home integrated intelligence tailored to their own needs.

3.2 Notions and Expectations of Human-Vehicle-Home Integrated Intelligence

While a comprehensive conceptualization of HVH integrated intelligence remains absent in the HCI community, this work does not seek to define it. Instead, through participants’ positive stories about their future, we identified three key values people associate with intelligent car-home connectivity that serves their needs: 1)

well-being in cross-space context, 2) continuums in activity and memory, and 3) enhancing interpersonal support and strengthening family bonds.

1) **Well-being in cross-space context:** Participants envisioned scenarios where intelligent and automated home-vehicle connectivity could enhance the quality of life. These included “perceptive car-home connections that locate items,” “pre-adjusted car temperatures based on travel habits,” and “energy-efficient bathing systems synced with the drive home”, which align with a series of well-being concerns emerging in home HCI such as convenience, sustainability, comfort, and leisure. Notably, participants emphasized the importance of interconnected processes across spaces, with the car facilitating seamless transitions. While some companies have already integrated home device control into car systems, how smart cars can truly foster cross-space well-being invites further attention and exploration.

Here, the scenarios envisioned by our participants highlight three key roles of cars: (1) *as an adapter that meets people’s needs in multiple contexts*: For example, P2 envisions a car that automatically plans a route to the store when informed by a smart fridge that the beer at home needs to be restocked. In this scenario, the car functions as an adapter, coordinating with other smart devices to enable a variety of intelligent services spanning multiple locations. (2) *as a meaningful altered medium for interaction*: While people commonly use smartphones or other smart home control panels for managing devices, they view the car’s connection with home devices as a meaningful alternative in certain contexts rather than an unnecessary complication. For example, a reminder on the car’s central display about a home door being left open might capture attention more effectively than a phone notification, and it is also considered safer to receive essential home information through voice assistants instead of grabbing a phone while driving. (3) *as multifunctional spaces beyond mobility*: In the envisioned scenarios, cars are reimagined as multifunctional smart spaces, such as a massage room, a power source for the home, or a portable office. P8 even posed a provocative question: “What could the car be when not in use?”—highlighting a shift in expectations from cars as mere transportation tools to vehicles integrated with diverse functionalities within HVH connectivity, a notion further supported by Bastian et al.’s [23] exploration of smart cars and in-car activities in the context of autonomous vehicles. While some of these imaginative visions of cars may seem beyond the scope of our discussion on car-home connectivity, we argue that it is crucial to rethink the scalability of smart cars’ functions within the vehicle-home integration ecosystem, especially considering the evolving roles of cars

driven by advancements in intelligent capabilities and autonomous driving technologies [27].

2) *Continuum in activity and data memory*: While HVH intelligent experiences imply increased interactions with a number of independent smart devices, participants emphasized the continuum in memory and usage among those devices in cars and homes. On the one hand, it refers to some scenes embodying the seamless use of smart services across spaces, such as watching a Netflix series in the car that was left unfinished at home. The motivation behind this idea may stem from people's expectation of a continuum of activities across spaces [29]. On the other hand, such expectation means some data revealing personal preferences and historical settings can be shared across different smart devices so that the integrated system feels like a singular "immersive life assistant" (P6). P6 even extended this continuum and interoperability to conversational agents in future hotels, expressing the desire to apply personal preferences to it to achieve a home-like experience. This raises the question: what if enhanced intelligence and connectivity allowed people to carry and apply their configurations across spaces, much like physically lifting and placing objects?

3) *enhancing interpersonal support and strengthening family bonds*: Our card-sorting results reveal a diverse range of stakeholders associated with the HVH system, including family members, friends, visitors, service providers, and even potential criminals. Given the inseparable nature of family and the home context, participants viewed the HVH integrated experience as a system designed to benefit other household members. P8 emphasized the importance of considering families when envisioning alternative future scenarios during the interview. So she invited her parents to collaboratively generate stories and canvases and discover disparities together. Among the many scenarios created, connected intelligence was seen as a potential bridge for linking family members, evolving into a force for mutual care. This means that when family members need assistance—such as in the event of a traffic accident or when an electric car needs charging—the system could push notifications to other members. This is significant as it signals a humanistic vision, one that is still lacking in the technology-dominated narratives of future HVH integration, and prompts reflection on how smart, connected technology could serve as links to enhance family well-being.

At last, we found that the referential connotation of "car" and "home" can extend beyond personal ownership, encompassing rented cars, shared vehicles, or connections to multiple residences and localized infrastructures. Participants envisioned scenarios such as lending vehicles to others or integrating cars with shared or secondary dwellings, highlighting new design considerations and challenges. As such, our work does not aim to provide specific dimensions that define the social concept and scope of "human, vehicle, and home." Rather than defining fixed dimensions for the social concepts of them, we advocate for recognizing the spectrum of this concept across diverse contexts and use cases. This broader perspective invites reflection on the communities involved, evolving definitions of "home," and, with advancements in smart mobility, what might even be considered a "car."

3.3 Concerns and Challenges in Integrated Smart Experiences

Participants prominently raised various concerns regarding a more integrated and connected HVH smart experience. Some stemmed from intentionally problematized scenarios, while others emerged from cautiously optimistic future visions. We identified five primary concerns, outlining their origins and the potential challenges they pose to the development of a more integrated smart experience while also proposing design implications to address these issues.

1) **Amplified Privacy and Security Concerns**: While the pursuit of a seamless connected smart experience inevitably entails accepting the increased capture, upload, and sharing of personal information, however, the consequence of privacy exposure and data leaks could be amplified when the boundaries between cars and homes are dissolved. P4 reflected: "If my phone's information were stolen, that might be tolerable, but if my family members are at home, could someone remotely control my car or even use the smart kitchen to ignite gas—actions that are much more dangerous?" This is a good example of showing the expanded risk that a security breach from one family member or in one device could place others at unpredictable risk, potentially eroding trust in interconnected cars, home devices, and smartphones. As noted by Coppola and Morisio [9], connected cars face security risks such as remote control of critical functions, data breaches, and vehicle theft due to vulnerabilities in sensors, communication modules, and control systems. In HVH integrated systems, these risks expand further, as the deep interconnectivity between vehicles and smart homes could allow hackers to compromise both domains simultaneously. This cross-space vulnerability highlights the urgent need for robust, multi-layered security measures to protect user privacy and ensure safety across interconnected environments.

Privacy risks in cars, like passenger turnover and location tracking, challenge the perception of cars as private spaces. P6 illustrated this concern by referencing the French term "chauffeur" for the ideal in-car assistant: "I would want it to be discreet, like a professional driver—hearing private conversations but pretending not to." Additionally, the car's mobility makes it more likely to expose individuals' current locations and intentions, yet this information is not expected to be shared with family members at all times. Balancing privacy concerns and ensuring passengers' awareness and control over data collection remain critical challenges in designing vehicle-home connectivity. Therefore, we call for designers and researchers to prioritize a series of user-centric issues related to informed consent and control. This includes clearly informing users about what information is being collected, where it is collected, who it is transmitted to, and how it is transmitted, as well as providing users with granular control over data sharing and the ability to customize privacy settings based on their preferences. Additionally, designing "fuse" mechanisms to limit remote control or functionality in the event of data breaches or unauthorized access is essential to ensure that losses are contained and do not spread.

2) **Shared Usage**: While the vision of HVH integration focuses on empowering all household members, participants highlighted challenges in shared use among individuals with varying technological literacy. P1, a frequent buyer of smart home devices, shared: "It's a hassle living with family; you have to teach them how to

use it because kids and elderly people struggle with these things.” Prior research [33] emphasized addressing family dynamics to ensure the contestability of every member in using home automation effectively. However, as controlling a car is less accessible, some individuals may have limited contestability in managing the HVH system. Conflicts over device control could also arise when alternative methods for managing the home or car are introduced. Also, Guests may affect residents’ experiences, highlighting the need for “guest modes” in shared systems. As such, we emphasize the importance of designing for contestability in shared usage, ensuring that different individuals are “granted equal power to understand, shape, and challenge decision-making” [33] about how HVH systems should be used and configured. This approach aims to avoid marginalizing those with insufficient technological literacy, fostering inclusivity and equitable access to the benefits of HVH integration.

3) **Critical Safety Concerns:** Safety concerns in integrated Intelligence experience are primarily highlighted in driving-related scenarios, as the introduction of the car amplifies the consequences of system disruptions and errors. Participants emphasized the necessity of considering from a safety perspective on how people could control their home while in the car and conversely which devices have the ability to control in-car features. Additionally, unexpected interruptions in connectivity could cause panic among drivers, as P7 noted: “If my connection between the car and home suddenly cuts off, I would find it more frightening than not having such the ability at all because you don’t know what’s happening.” This issue may be particularly pronounced in underdeveloped areas, or some households with poor network and power infrastructure, where unstable connections could easily lead to non-use [6] at times when safety is people’s primary concern.

While previous research on connected cars has identified several factors that impact driving safety, particularly focusing on the design of In-Vehicle Infotainment (IVI) systems and the integration of smartphones with car dashboards [15, 26, 34], our findings suggest that future research must also consider the broader implications of interconnected ecosystems, including the emotional and cognitive effects of system unpredictability and the need for robust fail-safe mechanisms to ensure user trust and safety.

To address these safety concerns, we propose two key design implications: *Examine the impact of in-car functions on driving safety*, particularly during the transition period when autonomous driving capabilities are not yet fully mature. This includes evaluating whether certain home-control features (e.g., adjusting thermostats or lighting) could distract drivers or compromise safety. *Ensure clear, non-invasive and immediate communication of system status* to users to prevent panic during unexpected connectivity interruptions. It is vital to offer this transparency to users to understand whether the system is functioning normally.

4) **Human Agency and Smart Technology as Agents:** While people look forward to more automated and intelligent technologies taking over certain decision-making tasks—such as adjusting devices based on their daily routines—they still argue for the necessity of human decision-making in some processes to avoid ill-equipped intelligence. P3 showcased a similar concern with a scenario where an automated home-vehicle system reduces energy consumption: “If I do this action every Friday, it may automatically arrange it for me the next Friday. But if something changes this Friday, I’m afraid

it will still arrange it, which defeats my purpose of saving energy and could cause some trouble.” On the other hand, participants also noted that as the car’s role expands, its absence—whether not in use or not physically present—may prevent it from fulfilling its responsibilities. As such, we argue that the insistence on human agency [19, 22] reflects a deeper concern about the inability of intelligent systems to adapt to the dynamic complexities of daily life. Our design implications emphasize the need to address these concerns by ensuring that HVH systems are flexible, context-aware, and capable of supporting—rather than replacing—human decision-making in diverse and evolving scenarios.

5) **Mismatch in High Intelligence:** Although most envisioned scenarios are set in a utopian or dystopian world perfectly equipped with intelligent systems, a more realistic concern is that, due to regional development disparities, many places may not be able to support smart interactions as expected. This could lead to new pragmatic challenges when people transition from a highly intelligent environment to one that is relatively less smart. P1 illustrated this challenge with a scenario: “In your previous home, everything was smart, right? But when you move to a new place, how do you adapt to the change in environment? You might imagine wanting a smart elevator to be ready, but you might find the new place doesn’t have a smart elevator.” How can we bridge the gap in infrastructure support for smart services and enable high-intelligence ecosystems to overcome these obstacles? We emphasize the importance of addressing this issue, as it will be a pragmatic challenge that accompanies the development of more intelligent HVH systems in the long term.

4 Conclusion

This study employed scenario envisioning with HVH probes to explore participants’ understandings, expectations, and concerns regarding intelligent connectivity between vehicles and homes. We introduce a probe-based methodology that empowers participants to connect their imaginative scenarios with real-life contexts, values, and routines. This approach not only facilitates the co-creation of future-oriented design ideas but also ensures that these ideas are grounded in the diverse and dynamic realities of everyday life, serving as a model for engaging non-technical users in envisioning and critiquing emerging technologies.

We provide the first empirical exploration of HVH systems from a human-centered perspective, uncovering key opportunities such as enhancing cross-space well-being, fostering seamless activity and memory continuums, and strengthening interpersonal support, particularly within families. These findings challenge the prevailing technology-centric narratives and highlight the importance of designing HVH systems that prioritize human needs and values. Additionally, our study reveals the evolving roles of cars within HVH ecosystems, envisioning them not only as mobility tools but also as adapters that meet diverse contextual needs, meaningful mediums for interaction, and multifunctional spaces that extend beyond transportation. This reimagining of the car’s role underscores the potential for vehicles to become integral components of a broader intelligent ecosystem. Furthermore, we identify critical socio-technical challenges in HVH integration, including amplified privacy and security risks, shared usage conflicts among individuals with varying technological literacy, safety concerns, and

mismatches in technological infrastructure across regions. These insights offer actionable guidance for addressing ethical, usability, and inclusivity issues in the development of HVH systems.

Addressing these challenges necessitates a commitment to accountability, fostering collaboration among diverse stakeholders, and implementing robust safeguards to cultivate trust, ensure security, and promote inclusivity. While this study provides valuable insights into HVH integration, it is important to acknowledge that the participants were all from China, which may introduce cultural nuances that shape their expectations and concerns. For instance, the emphasis on family-centric values and intergenerational support in the envisioned scenarios reflects cultural norms prevalent in the region. Future work could expand the participant pool to include individuals from diverse cultural backgrounds, offering a broader perspective on how cultural attitudes influence perceptions of HVH systems. This research lays the groundwork for designing HVH systems that move beyond technical advancement to embrace the diversity of communities and lifestyles, ultimately prioritizing human well-being and societal benefit.

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A Appendix: Workbook and Thematic Card Design

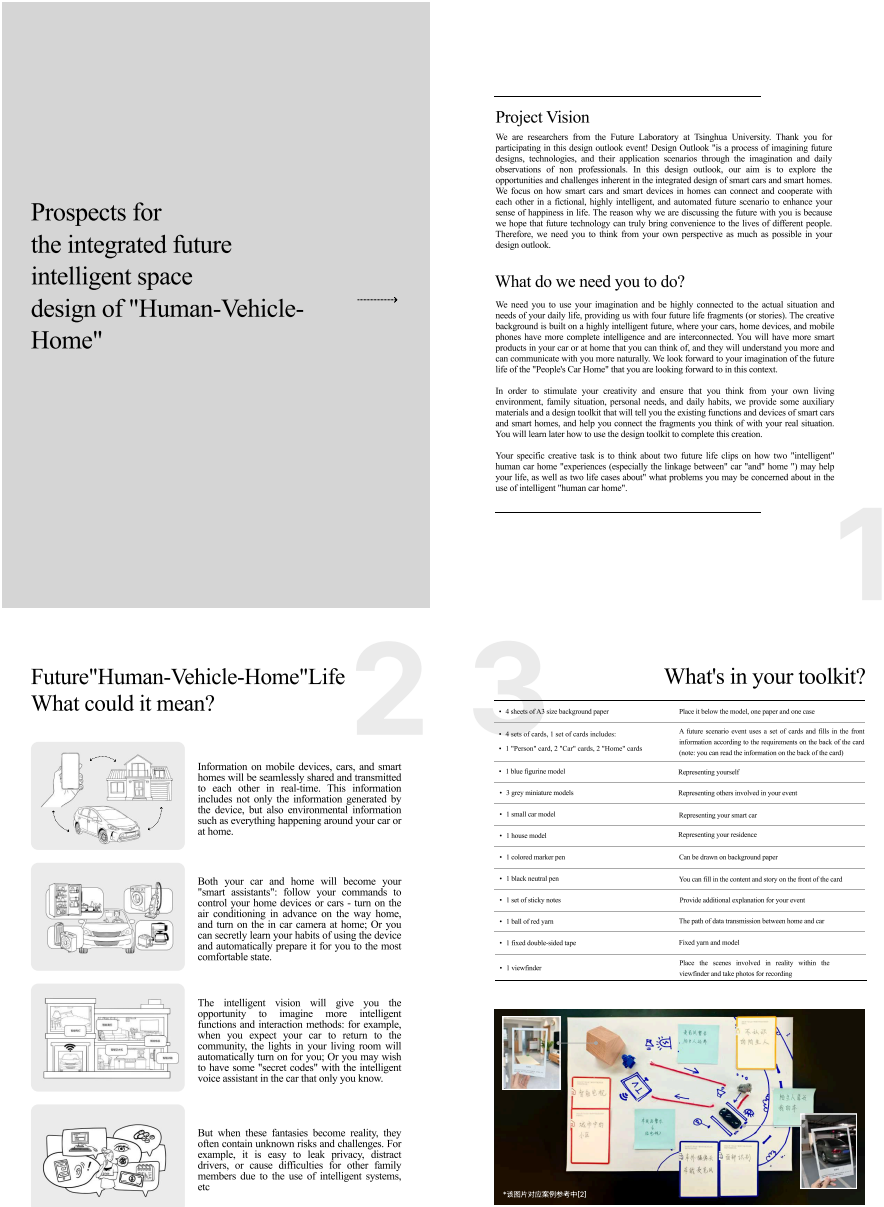


Figure 3: Contents of the Human-Vehicle-Home (HVH) probe package: (1) project vision (2) future “human-vehicle-home” life overview (3) toolkit overview

How to use a viewfinder?

In order to better connect with your actual daily scenarios and the fragments you have conceived, while allowing us to have a more concrete understanding of your specific situation regarding people, cars, and homes. We provide a viewfinder, where you need to place real-life scenes (such as your home environment, interior layout, or car related scenes) and capture them. Finally, you need to send the captured images to the contact person.



Case reference

We have provided some examples of previous participants for your reference. We want you to start the story from your daily habits, experiences, preferences, family and living situation, occupation and background. Look forward to learning about your personal ideas and needs for the intelligent life of the future, and get your own story.

Home Vehicle Home

How could the intelligent "Human-Vehicle-Home" experience potentially help improve your life?

- [1] I hope that my car and home devices can be connected with a single brain, capable of remembering my daily routine tasks. For example, my child goes to elementary school, and every morning, I need to get her ready and then take her to school. After waking up, I would use some smart home devices like smart curtains and a robot vacuum, which could automatically adjust according to my daily habits, without me needing to manually set them up. When I get in the car, if I forgot to close the door or turn off other devices at home, the car can notify me via voice or even show them for me. My child can watch English learning cartoons in the backseat, and any unfinished content at home can be directly resumed in the car.
- [2] I would hope that cars could make progress in terms of safety. For instance, if I usually park my car outside, and the car detects a stranger lingering around it for ten minutes without being on my "trusted list", I would want the car to send me an alert. At that moment, I might be at home watching TV, so I'd like the important alerts to show up directly on my TV. From there, I could use the car's exterior cameras to see what the person is doing and check the surveillance footage. I'd even hope to be able to activate the car's interphone to ask the person to stay away.
- [3] I would like the car to become a personal space that allows me to relax and focus on my health. For example, while driving, it could monitor my health and accumulate data every day. If there are any abnormalities, it could send a report to my phone and even predict any potential health risks. If the air quality in the car isn't ideal, it could automatically open the windows for ventilation or diffuse some aromatherapy. Additionally, there could be entertainment features so that even when I'm away from home, I can still access cloud services from my home devices and enjoy them in the car.

What issues would you be concerned about in the use of an intelligent "Human-Vehicle-Home" system?

- [4] Although the car might be able to understand some of my needs from what I say and automatically adjust certain settings, this could make me feel that my privacy is being violated. For example, I might walk into my house and notice that the temperature is much higher than usual, only to realize it's because I gave a colleague a ride after work, and during the ride, they mentioned feeling cold, prompting me to raise the car's temperature. As a result, the house's air conditioning was also automatically adjusted to a higher setting. On one hand, it feels like my entire life and privacy are being monitored, and on the other hand, should the system have checked with me before adjusting the temperature? Also, sometimes when I'm giving a ride to my boss, certain conversations being automatically recorded by the system could be inappropriate. Shouldn't the system also consider the rights of others involved, especially regarding consent?
- [5] I usually enjoy buying these smart devices, but our family is a large one with many members, and we often have guests. I'm a bit worried that the elderly in our family might find it difficult to use these devices. Also, my husband and I both drive this car, and since our settings might be different, I'm concerned they might conflict with each other.
- [6] My wife and I drive the same car, and both of our phones are connected to the car's customizable navigation system. However, since we can both access the car's sensors, location data, it could lead to privacy issues. For example, sometimes I like to sit in the car for a while before heading inside, but she can see my location and notice that I'm already downstairs...

Please share your story with us!

Please record the four scenarios you have envisioned in writing.

Story 1 (Better Life Experience) :

Story 2 (Better Life Experience) :

Please share your story with us!

Please record the four scenarios you have envisioned in writing.

Story 3 (Worries and Concerns) :

Story 4 (Worries and Concerns) :

Figure 4: Contents of the Human-Vehicle-Home (HVH) probe package: (4) guide for using the framing device (5) case reference and (6, 7) templates for story sharing.

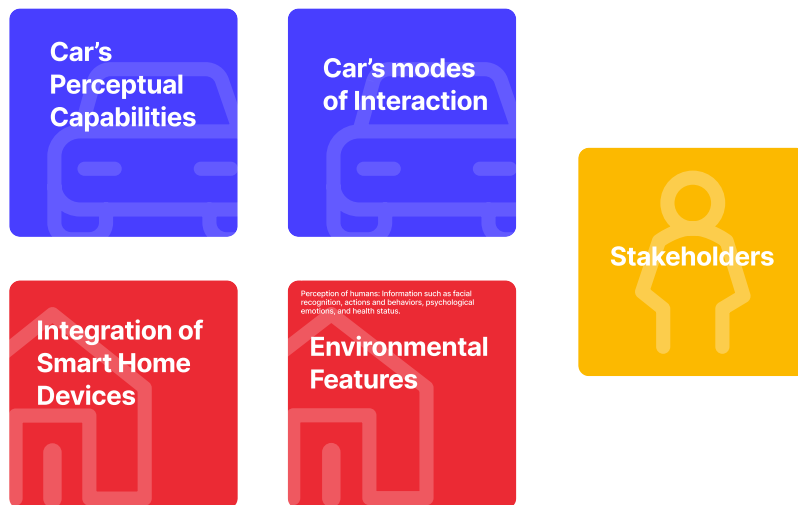


Figure 5: The package also includes thematic cards, covering five categories: car's perceptual capabilities, car's modes of interaction, integration of smart home devices, environmental features, and stakeholders, designed to guide participants in their scenario creation process.